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Primary and Secondary (Nongenetic) Causes of Focal and Segmental Glomerulosclerosis

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INTRODUCTION

Focal segmental glomerulosclerosis (FSGS) is a histologic pattern of glomerular injury that represents the final common pathway of podocyte injury and depletion. FSGS may be primary (idiopathic) or secondary to diverse causes[1-3] (see also Chapter 20). Early in the disease process, the pattern of glomerulosclerosis is focal, involving a minority of glomeruli, and segmental, involving a portion of the glomerular tuft.[4] As FSGS progresses, more diffuse and global glomerulosclerosis evolves.[4] Although it accounts for only a small proportion of nephrotic syndrome in young children, FSGS represents as many as 35% of cases of primary nephrotic syndrome in adults and is a major cause of end-stage kidney disease (ESKD) in the United States.[5]

FSGS can be caused by a diverse set of pathogenetic mechanisms, some of which manifest as particular histologic subtypes of disease. Although primary (idiopathic) FSGS is potentially treatable and curable in many patients, the optimal type and duration of immunosuppressive, as well as adjunctive therapy, remain controversial. For secondary FSGS, effective therapies exist to slow or modify the disease course (see Chapter 82).

ETIOLOGY AND PATHOGENESIS

FSGS is a pattern of glomerular injury that represents a common final pathway of podocyte depletion. Primary FSGS underlies a subset of idiopathic nephrotic syndrome and is thought to be mediated by circulating permeability factors.[6-8] Distinct causes of secondary FSGS (Box 19.1) include genetic variants in podocyte and glomerular basement membrane components (see Chapter 20), viral infections, drug toxicity, maladaptive responses to a reduced number of functioning nephrons, and hemodynamic stress placed on an initially normal nephron population. In all forms of FSGS, injury directed at or inherent to the podocyte mediates altered cell signaling and reorganization of the actin cytoskeleton resulting in foot process effacement, and eventually podocyte depletion though detachment and apoptosis.[1,9] Podocytes are terminally differentiated, postmitotic cells with limited hypertrophic capacity. Furthermore, regeneration mediated by resident podocyte progenitor cells along Bowman's capsule is inefficient. Therefore, stress placed on the remaining podocytes may lead to local propagation of damage (see Chapter 81). Injury to podocytes may spread to adjacent podocytes by reduction in supportive factors such as nephrin signaling, increased toxic factors such as angiotensin II (Ang II), or mechanical strain on remnant podocytes.[9,10] Cell-to-cell spread of podocyte injury until the entire glomerular lobule is captured could explain the segmental nature of the sclerosing lesions.[10] Repeated and diverse stresses to the podocyte could explain the FSGS pattern of injury in age-related nephrosclerosis and glomerular senescence.[1,9]

CLASSIFICATION OF FOCAL SEGMENTAL GLOMERULOSCLEROSIS

The terms *primary* and *idiopathic* FSGS have been used interchangeably, leading to confusion. Thus, we avoid the term *idiopathic* in this section. The following definitions are suggested by the KDIGO Guideline.[11]

Primary Focal Segmental Glomerulosclerosis

Primary FSGS is diagnosed when the nephrotic syndrome is present (in particular, the presence of hypoalbuminemia) and diffuse podocyte foot process effacement is observed on kidney biopsy without evidence for other causes of FSGS.[11] Primary FSGS, together with minimal change disease (MCD; see Chapter 18), are the main causes of idiopathic nephrotic syndrome and are collectively referred to as *primary podocytopathies*. Compared with MCD, primary FSGS is associated with a higher likelihood of steroid resistance and progression to CKD, leading most experts to conclude that they are separate diseases. Despite these clinical differences, MCD and primary FSGS may represent a disease spectrum in which the severity and duration of podocyte injury, inherent susceptibility of the podocyte to injury, and comorbid conditions all influence the disease phenotype.[12] Indeed, follow-up kidney biopsies from patients with refractory nephrotic syndrome suggest that FSGS may evolve from an initial MCD pattern.[13] In some instances this may be attributed to sampling variability. However, sequential biopsy samples of kidney allografts show that recurrent FSGS may pass through an early stage of podocytopathy with MCD-like pathology.[14] Animal models of persistent nephrotic syndrome also show progression from an initial MCD-like phase to FSGS.[15]

Both MCD and primary FSGS are thought to be mediated by as-yet unidentified circulating permeability factors.[6,16] Alterations in glomerular capillary wall permeability may occur in response to circulating "humoral" substances that act on the podocyte to promote foot process effacement. Circulating permeability factors that enhance in vitro permeability of glomeruli to albumin have been found in some FSGS patients. An in vitro assay of cultured podocytes has been used to detect the presence of circulating permeability factors.[17] Some patients with recurrent FSGS after transplantation achieve remission of nephrotic syndrome after plasma exchange or use of a protein A adsorption column, supporting the role of a circulating factor[8,18] (see Chapter 113). However, none of several candidate proteins, including suPAR, B7-1 (also known as CD-80), or cardiotrophin-like cytokine 1 (CLC1), a member of the interleukin (IL)-6 family, has been consistently shown to be the permeability factor in human FSGS.[19]

Glomerular hypertrophy (or glomerulomegaly) may identify children with MCD at risk for development of FSGS. In early idiopathic FSGS and in many secondary forms of FSGS, such as obesity-related FSGS, there is initially glomerular hypertrophy and high glomerular filtration rate (GFR) associated with glomerular hypertension.[20] Similarly, in secondary forms of FSGS with reduced nephron numbers, maladaptive

BOX 19.1 Etiologic Classification of Focal Segmental Glomerulosclerosis

Primary
- Presumably mediated by circulating/permeability factor(s)

Genetic Causes
- (See Chapter 20)

Secondary

Virus Associated
- HIV-1 (HIV-associated nephropathy)
- SARS-CoV-2 (COVID-19 associated nephropathy)
- Parvovirus B19
- CMV
- EBV

Drug Induced
- Interferon
- Pamidronate
- Sirolimus
- Anabolic steroids

Adaptive Changes to Glomerular Hyperfiltration
Reduced Kidney Mass
- Oligomeganephronia
- Very low birth weight (correlating with low nephron endowment)
- Unilateral kidney agenesis
- Kidney dysplasia
- Reflux nephropathy
- Sequela to cortical necrosis
- Surgical kidney ablation
- Any advanced kidney disease with reduction in functioning nephrons

Initially Normal Kidney Mass
- Hypertension
- Atheroemboli or other acute vasoocclusive processes
- Obesity
- Bodybuilders
- Cyanotic congenital heart disease
- Sickle cell anemia

Uncertain Cause

CMV, Cytomegalovirus; *EBV,* Epstein-Barr virus; *FSGS,* focal segmental glomerulosclerosis; *HIV,* human immunodeficiency virus; *SARS-CoV-2,* severe acute respiratory syndrome coronavirus 2.

hemodynamic alterations may be associated with glomerular hyperfiltration. Intraglomerular coagulation and abnormalities of lipid metabolism may also contribute to glomerulosclerosis in these patients (see Chapter 81). Although immunoglobulin (Ig) M and C3 are regarded as nonspecifically trapped in areas of sclerosis, they may contribute to glomerular injury through activating complement and binding antigens.[21,22] Recent studies document both reparative and sclerosing roles for parietal epithelial cells in FSGS. In vivo microscopy has illustrated that parietal cells can rapidly cover sites of podocyte denudation and depletion in FSGS but make a limited contribution to podocyte regeneration.[23,24] Parietal cells may also directly contribute to glomerulosclerosis by migrating onto the glomerular tufts and synthesizing extracellular matrix proteins.[25]

Genetic Focal Segmental Glomerulosclerosis

Genetic and familial forms of FSGS are covered in detail in Chapter 20. Many cases of apparently primary FSGS may harbor unrecognized variants or polymorphisms in podocyte genes. Genetic predisposition may underlie the susceptibility to second hits, whereby viral factors, immune stimuli, and other insults lead to the initiation or acceleration of disease. For example, *APOL1* gene variants predispose to FSGS, as well as HIV nephropathy, chronic hypertensive nephrosclerosis, and lupus nephritis among African Americans.[26] G1 and G2 variants in *APOL1* protect against infection by *Trypanosoma brucei*, the parasite that causes African sleeping sickness. Similar to the gene for sickle cell disease, which confers protection against malaria, the G1 and G2 variants became prevalent because they protect against infection. Although *APOL1* is expressed by glomerular endothelial cells and podocytes, it is not entirely clear how sequence variations in *APOL1* cause glomerulosclerosis. Various mechanisms have been proposed including increased podocyte membrane pores promoting intracellular potassium depletion and induction of stress-activated protein kinases[27] and impairment of podocyte endocytic functions and autophagy.[28]

Secondary Focal Segmental Glomerulosclerosis

When an FSGS lesion is found together with a process associated with FSGS, we refer to this as secondary FSGS.[11] The known/presumptive etiologies of secondary FSGS include viral infections, drug toxicities, and maladaptive responses to glomerular hyperfiltration and are listed in Box 19.1.

Virus-Associated Focal Segmental Glomerulosclerosis

Viral infections (particularly HIV) can cause podocytopathies, particularly collapsing FSGS (see Chapter 58). Other potential viral triggers include parvovirus B-19, Epstein-Barr virus (EBV), and cytomegalovirus (CMV) infections. During the COVID-19 pandemic, collapsing FSGS was linked to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection in patients of African ancestry.[29,30] The heightened cytokine release associated with viral infections may place patients with the high-risk *APOL1* genotype at risk of interferon-mediated podocyte injury, as can occur in other virally mediated, interferon therapy–induced, and HIV-associated forms of collapsing FSGS.[31,32]

Drug-Induced Focal Segmental Glomerulosclerosis

Several drugs and medications are associated with the FSGS phenotype, including heroin, lithium, pamidronate, sirolimus, calcineurin inhibitors (CNIs), tyrosine kinase inhibitors, and interferons α, β, and γ (Box 19.1). Heroin is associated with the nephrotic syndrome and FSGS (heroin nephropathy), although its incidence has diminished markedly in recent years. Pamidronate is a bisphosphonate used to prevent bone resorption in myeloma and metastatic tumors and is associated with collapsing FSGS and MCD.[33] Stabilization of kidney function and resolution of nephrotic syndrome may follow withdrawal of these medications. Long-term anabolic steroid use by bodybuilders is associated with FSGS,[34] although this observation could be confounded by use of high-protein diets and other potential nephrotoxins such as growth hormone. Putative mechanisms of glomerular injury include direct toxic effects of anabolic steroids on glomerular cells, as well as adaptive responses to elevated lean body mass. In a small series, all patients who developed FSGS secondary to interferon therapy have been shown to carry double *APOL1* risk alleles and to activate a viral innate immunity program in podocytes upon interferon exposure, supporting a two-hit model of podocyte injury.[35]

Focal Segmental Glomerulosclerosis From Adaptive Changes to Glomerular Hyperfiltration

Many secondary forms of FSGS are mediated by adaptive structural-functional responses leading to podocyte detachment and subsequent sclerosis.[36] These adaptive forms include patients with both congenital and acquired reduction in the number of functioning nephrons, whereas other secondary forms are associated with hemodynamic stress placed on an initially normal nephron population (Box 19.1). Obesity-related glomerulopathy is increasingly common worldwide and may be associated with metabolic syndrome, including hypertension, diabetes, and hyperlipidemia.[37] Low birth weight associated with prematurity and reduced nephron endowment also may lead to glomerular hypertrophy, with secondary FSGS developing in adolescence or adulthood.[38] Biopsy specimens with secondary adaptive FSGS typically show glomerulomegaly and perihilar lesions of segmental sclerosis and hyalinosis. These conditions resemble experimental models of kidney ablation in which the surgical reduction in kidney mass causes functional hypertrophy of remnant nephrons with increased glomerular plasma flows and pressures. Although these changes are initially "adaptive," the resultant hyperfiltration and increased glomerular pressure become "maladaptive" and serve as mechanisms for progressive glomerular damage.[36]

Focal Segmental Glomerulosclerosis From Uncertain Cause

FSGS can also occur without a genetic or identifiable secondary cause, in the absence of nephrotic syndrome (NS), and without diffuse foot process effacement on electron microscopy of the kidney biopsy. This form of FSGS is distinct from primary FSGS based on its clinical and histologic manifestations. The KDIGO Guideline proposes calling this entity FSGS of unknown cause.[11] Patients with FSGS of unknown cause may have secondary or genetic forms of FSGS that have not yet been elucidated.

EPIDEMIOLOGY

There is substantial variation in the reported frequency of primary glomerular diseases between different geographic and among racial populations.[39,40] The reasons for these differences may be multifactorial (e.g., bias in referral patterns and who undergoes biopsy, environmental or racial predispositions such as the presence of *APOL1* genetic variants in people of African descent). The prevalence of FSGS varies across countries. In a study from China, FSGS was diagnosed in 2.45% of biopsies.[41] In a Danish kidney biopsy registry study, an incidence of 1.5 to 5.7 patients per million/year was reported.[42] In a single-center study from Germany, FSGS accounted for 6.1% of biopsies, and its frequency rose 3.9-fold over a 24-year period beginning 1990.[43] An analysis of the prevalence of ESKD in the United States caused by FSGS during a 21-year period showed an increase from 0.2% in 1980 to 2.3% in 2000, and FSGS was the most common primary glomerular disease leading to ESKD.[5] Although some changes in prevalence may relate to changes in biopsy practice or disease classification, the true frequency of FSGS has likely increased over time.

Primary FSGS is slightly more common in males than females, and the incidence of ESKD secondary to FSGS in males of all races is 1.5 to 2 times higher than in females. The incidence in both children and adults is higher in African Americans than in Whites and those of Asian descent.[1,44] This most likely relates to the presence of G1 and G2 variants of the apolipoprotein L1 genes in African Americans.[26]

CLINICAL MANIFESTATIONS

Patients with primary FSGS typically present with the full nephrotic syndrome (especially hypoalbuminemia).[11,45] Secondary forms of FSGS associated with hyperfiltration typically have lower levels of proteinuria, and many such patients have subnephrotic proteinuria and normal serum albumin concentration.[37,46] Patients with genetic FSGS may have variable degrees of proteinuria and serum albumin levels.[46]

Hypertension is found in 30% to 65% of children and adults with FSGS at diagnosis. Microhematuria is found in 30% to 75% of these patients, and decreased GFR is noted at presentation in 20% to 50%.[1] Daily urinary protein excretion ranges from less than 1 to more than 30 g/day. Proteinuria is typically nonselective. Complement levels and other serologic test results are normal. Occasional patients will have glycosuria, aminoaciduria, phosphaturia, or a concentrating defect indicating functional tubular damage and glomerular injury.

Different histologic patterns of FSGS may display unique clinical features.[47] The clinical presentation of patients with the tip variant of FSGS resembles MCD.[48] They often present with an abrupt clinical onset of the full nephrotic syndrome (almost 90%), shorter time course from onset to kidney biopsy, more severe proteinuria, and less chronic tubulointerstitial disease than in FSGS not otherwise specified (NOS).[48] The cellular variant also typically manifests with greater proteinuria and higher incidence of nephrotic syndrome than FSGS NOS.[49] The collapsing variant usually manifests with greater proteinuria, severe nephrotic syndrome, and lower GFR.[47,50]

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

Except for genetic forms of FSGS, a firm diagnosis of FSGS requires a kidney biopsy. Tests for permeability factors are neither reliable nor available in routine clinical practice. In children with FSGS, most of whom present with nephrotic syndrome, the major differential is between MCD and other variants of corticosteroid-resistant nephrotic syndrome. In adults with subnephrotic proteinuria, the differential includes almost all glomerular diseases without positive serologic results. In adults with nephrotic syndrome, MN and MCD may present in an identical manner, and only a kidney biopsy will clarify the diagnosis (except for MN patients, in whom the presence of M-type phospholipase A2 receptor [PLA2R] antibody and possibly other autoantibodies may provide the diagnosis without a biopsy; see Chapter 21). Focal sclerosing lesions caused by other glomerulopathies (e.g., segmental scarring from chronic glomerulonephritis) must be excluded pathologically. Moreover, because the defining glomerular lesion of FSGS is focal and may be confined to deeper juxtamedullary glomeruli early in the disease, it may not be sampled on kidney biopsy. A large sample of more than 20 glomeruli for light microscopy or serial sections of the biopsy core may increase the likelihood of identifying the diagnostic segmental lesions.

Even after the diagnosis of FSGS is established, the primary form must be distinguished from secondary forms by careful clinicopathologic correlation (Box 19.2). In general, many forms of adaptive FSGS have lower levels of proteinuria than primary FSGS, a lower incidence of hypoalbuminemia, and, on biopsy, lesser degrees of foot process effacement.[11,46] In patients younger than 25 years and especially in those with a family history of FSGS, genetic screening for variants in podocin, TRPC6, α-actinin-4, inverted formin 2, or other podocyte genes may be useful (see Chapter 20). Similarly, in young adults with sporadic corticosteroid-resistant FSGS, genetic testing may yield pathogenic variants, albeit in a lower proportion compared with children.[51,52]

PATHOLOGY

The pathologic manifestations of FSGS are heterogeneous.[1,3] A hierarchical classification of FSGS by histologic variants (Table 19.1) can be applied to both primary and secondary forms of FSGS (see Box 19.1).[47] This working classification has been applied successfully to retrospective and prospective biopsy series. Other, more controversial histologic variants of FSGS include FSGS with diffuse mesangial hypercellularity and C1q nephropathy (see Chapter 29). Some think these are distinct disease entities, whereas others think they are merely subgroups of FSGS.[53,54]

BOX 19.2 Definition of Remission, Relapse, Resistance, and Therapy Dependence for Focal Segmental Glomerulosclerosis

Complete Remission
Reduction of proteinuria to <0.3 g/day or urine PCR <300 mg/g (or <30 mg/mmol), stable serum creatinine, and serum albumin >3.5 g/dL (or 35 g/L)

Partial Remission
Reduction of proteinuria to 0.3–3.5 g/day or urine PCR 300–3500 mg/g (or 30–350 mg/mmol) and a decrease >50% from baseline

Relapse
Proteinuria >3.5 g/day or urine PCR >3500 mg/g (or 350 mg/mmol) after complete remission has been achieved or an increase in proteinuria by >50% during partial remission

Steroid-Resistant
Persistence of proteinuria >3.5 g/day or urine PCR >3500 mg/g (or 350 mg/mmol) with <50% reduction from baseline despite prednisone 1 mg/kg/day or 2 mg/kg every other day for at least 16 weeks

Steroid-Dependent
Relapse occurring during or within 2 weeks of completing glucocorticoid therapy
CNI-resistant FSGS
Persistence of proteinuria >3.5 g/day or urine PCR >3500 mg/g (or 350 mg/mmol) with <50% reduction from baseline despite cyclosporine treatment at trough levels of 100–175 ng/mL or tacrolimus treatment at trough levels of 5–10 ng/mL for >6 months

CNI-Dependent
Relapse occurring during or within 2 weeks of completing cyclosporine or tacrolimus therapy for >12 months

CNI, Calcineurin inhibitor; *FSGS,* focal segmental glomerulosclerosis; *PCR,* polymerase chain reaction.
Modified from KDIGO GN Guideline. *Kidney Int.* 2021;100(4S):S1–S276.

Classic Focal Segmental Glomerulosclerosis Not Otherwise Specified

FSGS NOS requires exclusion of the more specific subtypes described later. This is the most frequent variant and may occur in primary or secondary forms of FSGS, including genetic forms. It is defined by segmental obliteration of the glomerular tuft by matrix accumulation, which may be accompanied by hyalinosis and/or endocapillary foam cells (Fig. 19.1).[47] Adhesions or synechiae to Bowman's capsule are common, and overlying visceral epithelial cells often appear swollen and form a cellular "cap" over the sclerotic segment. Glomerular lobules unaffected by segmental sclerosis appear normal on light microscopy, except for mild podocyte swelling. Tubular atrophy and interstitial fibrosis are commensurate with the degree of glomerulosclerosis. Immunofluorescence (IF) typically reveals focal and segmental granular deposition of IgM, C3, and more variably C1q in the distribution of the segmental glomerular sclerosis (Fig. 19.2). On electron microscopy (EM), segmental sclerotic lesions exhibit increased extracellular matrix (ECM), wrinkling and retraction of the glomerular basement membrane (GBM), accumulation of inframembranous hyaline, and resulting narrowing or occlusion of the glomerular capillary lumina. Electron-dense immune-type deposits should be absent and, if present, should raise suspicion for chronic glomerulonephritis. Overlying the segmental sclerosis, there is often podocyte detachment with parietal cell coverage (Fig. 19.3). The adjacent nonsclerotic glomerular capillaries show foot process effacement with variable microvillous transformation (slender projections resembling villi along the surface of podocytes).

Perihilar Variant of Focal Segmental Glomerulosclerosis

The perihilar variant is defined as hyalinosis and sclerosis at the vascular pole involving more than 50% of glomeruli with segmental lesions (Fig. 19.4).[47] This category requires exclusion of the cellular, tip, and collapsing variants. Although the perihilar variant may occur in primary FSGS, it is particularly frequent in secondary forms of FSGS mediated by adaptive structural-functional responses, accompanied by glomerular hypertrophy (glomerulomegaly) and relatively mild foot process effacement. In this setting, reflex dilation of the afferent arteriole and the greater filtration pressures at the proximal end of the

TABLE 19.1 Morphologic Variants of Focal Segmental Glomerulosclerosis : Columbia Classification

Variant	Defining Features	Clinical Features	Associations
Collapsing	Implosive retraction of capillaries with overlying GEC hyperplasia, severe tubular injury, tubular microcysts, severe FPE	Primary or secondary, severe nephrotic syndrome and AKI, Black racial predominance, worst prognosis	Viral infections (HIV, SARS-CoV-2, CMV), autoimmune SLE, drugs (pamidronate, INF), vasoocclusion, *APOL1*
Tip	Adhesion of tuft at tubular pole with foam cells, severe FPE	Usually primary, abrupt-onset nephrotic syndrome	Usually steroid responsive, favorable prognosis
Cellular	Expansile lesion with endocapillary hypercellularity (foam cells, leukocytes)	Usually primary	
Perihilar	Hyalinosis and sclerosis centered at vascular pole, glomerulomegaly, focal FPE	Usually secondary, adaptation to glomerular hyperfiltration	Obesity, HTN, low nephron number, sickle cell nephropathy
NOS	Does not meet criteria for any above variant	Most common variant, primary or secondary	

AKI, Acute kidney injury; *CMV,* cytomegalovirus; *FPE,* foot process effacement; *FSGS,* focal segmental glomerulosclerosis; *GEC,* glomerular epithelial cell; *HIV,* human immunodeficiency virus; *HTN,* hypertension; *INF,* interferon; *NOS,* not otherwise specified; *SARS-CoV-2,* severe acute respiratory syndrome coronavirus 2; *SLE,* systemic lupus erythematosus.

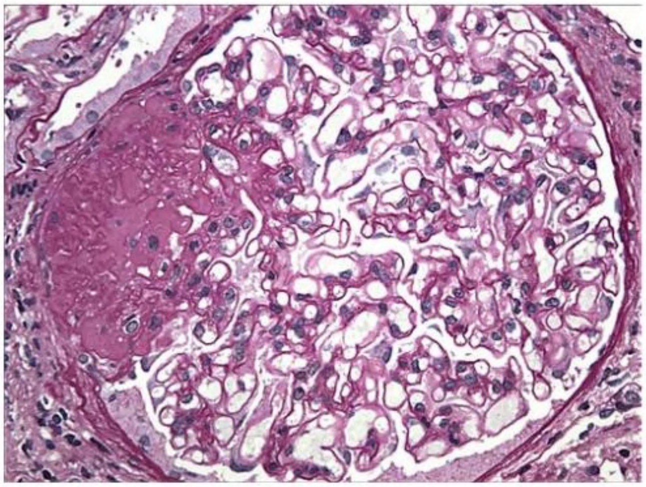

Fig. 19.1 **Focal Segmental Glomerulosclerosis Not Otherwise Specified.** A glomerulus with discrete lesion of segmental sclerosis involving a portion of the tuft characterized by accumulation of acellular matrix material, hyalinosis, and adhesion of the tuft to Bowman's capsule. (PAS; ×400.)

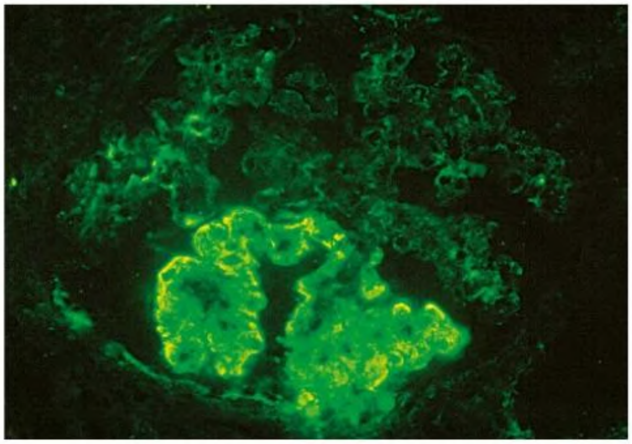

Fig. 19.2 **Focal Segmental Glomerulosclerosis Not Otherwise Specified.** The lesions of segmental sclerosis contain deposits of immunoglobulin M (IgM) corresponding to areas of increased matrix and hyalinosis. Weaker staining for IgM is also seen in the adjacent mesangium. (Immunofluorescence micrograph; ×400.)

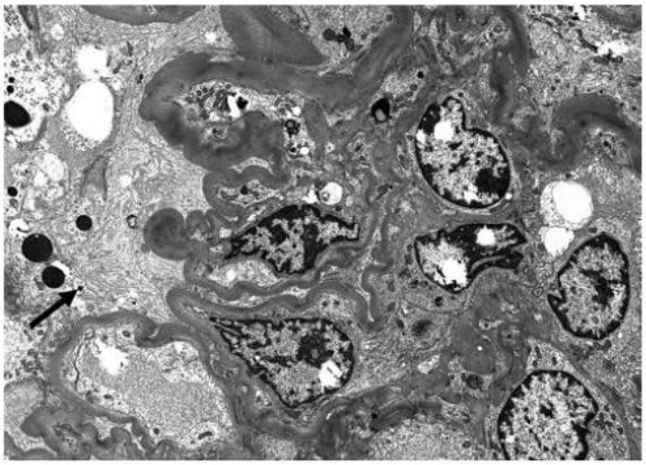

Fig. 19.3 **Focal Segmental Glomerulosclerosis Not Otherwise Specified.** Electron micrograph illustrates the lesion of segmental sclerosis with obliteration of the glomerular capillaries by increased extracellular matrix with wrinkled and retracted glomerular basement membranes. The overlying podocytes are detached *(arrow).* (×3000.)

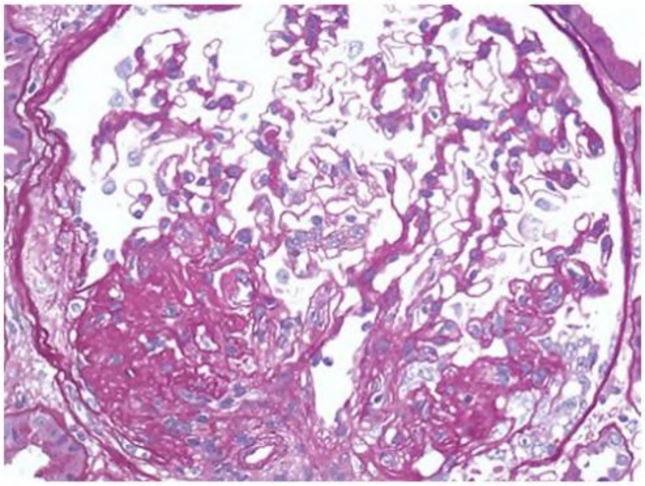

Fig. 19.4 **Focal Segmental Glomerulosclerosis (FSGS), Perihilar Variant.** A discrete lesion of segmental sclerosis and hyalinosis is located at the glomerular vascular pole (i.e., perihilar). The glomerulus is hypertrophied. The patient had secondary FSGS in the setting of severe obesity. (PAS; ×400.)

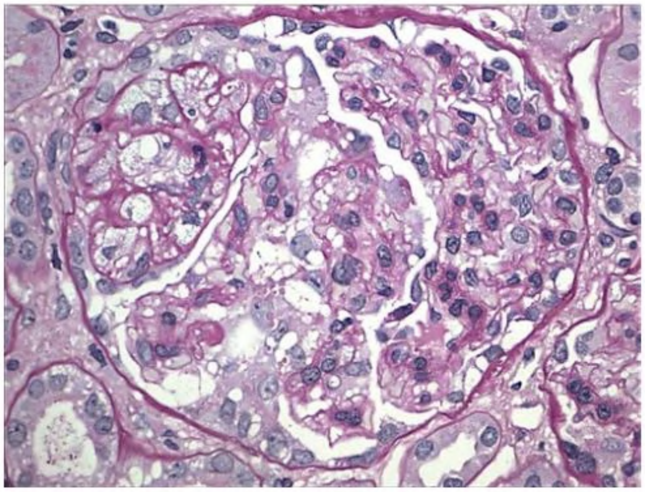

Fig. 19.5 **Focal Segmental Glomerulosclerosis, Cellular Variant.** The glomerular capillary lumina are segmentally occluded by endocapillary cells, including foam cells, infiltrating mononuclear leukocytes, and pyknotic debris. The findings mimic a proliferative glomerulonephritis because of the hypercellularity and absence of extracellular matrix material. There are hypertrophy and hyperplasia of the overlying visceral epithelial cells, some of which contain protein resorption droplets. (PAS; ×400.)

glomerular capillary bed may favor the development of lesions at the vascular pole.[20,36]

Cellular Variant of Focal Segmental Glomerulosclerosis

The cellular variant is characterized by focal and segmental endocapillary hypercellularity that may mimic a form of focal proliferative glomerulonephritis.[47,49] Glomerular capillaries are segmentally occluded by endocapillary hypercellularity, including foam cells, infiltrating leukocytes, karyorrhectic debris, and hyaline (Fig. 19.5). There is often hyperplasia of the visceral epithelial cells, which may appear swollen, sometimes forming pseudocrescents. Foot process effacement is typically severe. This variant requires that tip lesions and collapsing lesions be excluded. Cellular FSGS is thought to represent an early stage in the development of segmental lesions and is usually primary.

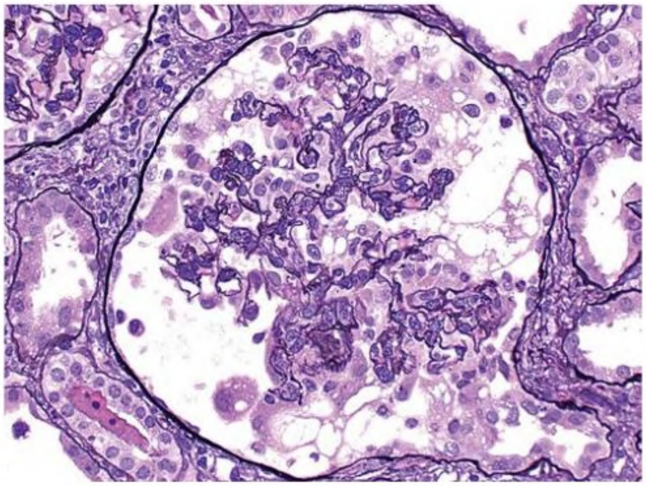

Fig. 19.6 Focal Segmental Glomerulosclerosis, Collapsing Variant. There is global implosive collapse of the glomerular tuft with obliteration of capillary lumina. The overlying visceral epithelial cells appear hypertrophied and hyperplastic, show prominent cytoplasmic vacuolization, and contain enlarged nuclei and nucleoli. There are no adhesions to Bowman's capsule. (Jones methenamine silver; ×400.)

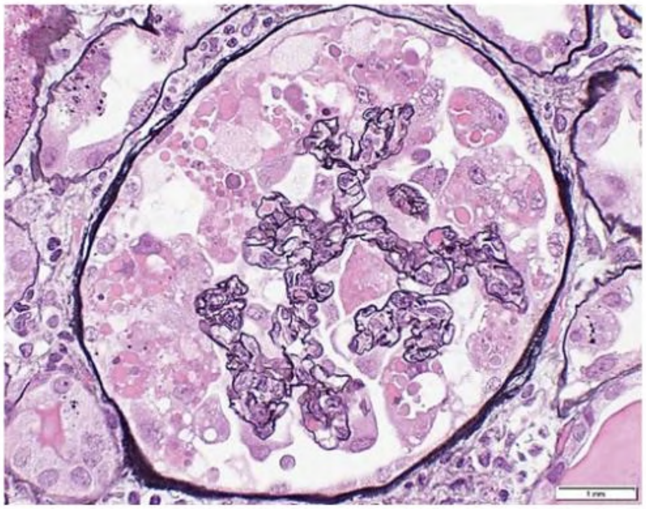

Fig. 19.7 Focal Segmental Glomerulosclerosis, Collapsing Variant. In this example, exuberant proliferation of glomerular epithelial cells forms a pseudocrescent that obliterates the urinary space. The hypertrophied and hyperplastic glomerular epithelial cells contain abundant intracytoplasmic protein droplets. The pseudocrescent lacks the spindle cell morphology, ruptures of Bowman's capsule, or pericellular matrix typically seen in true inflammatory crescents of parietal epithelial origin. (Jones methenamine silver; ×400.)

Collapsing Variant of Focal Segmental Glomerulosclerosis

The collapsing variant is defined by at least one glomerulus with segmental or global collapse and overlying hypertrophy and hyperplasia of visceral epithelial cells (Fig. 19.6). In these areas, there is occlusion of glomerular capillary lumina by implosive GBM wrinkling and collapse.[47,50] The collapsing lesion is more often global than segmental. Overlying visceral epithelial cells display hypertrophy and hyperplasia and express proliferation markers. These glomerular epithelial cells often contain prominent intracytoplasmic protein resorption droplets and may fill Bowman's space, forming pseudocrescents (Fig. 19.7). Attenuated collapsing lesions are characterized by retraction and solidification of the tuft, which is often capped by a layer of reactive

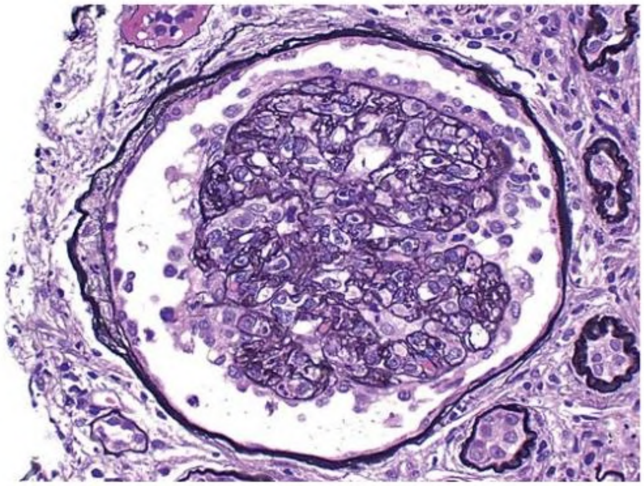

Fig. 19.8 Focal Segmental Glomerulosclerosis, Collapsing Variant. An attenuated lesion of collapsing focal segmental glomerulosclerosis shows global retraction and early solidification of the glomerular tuft, which is capped by reactive-appearing glomerular epithelial cells. (Jones methenamine silver; ×400.)

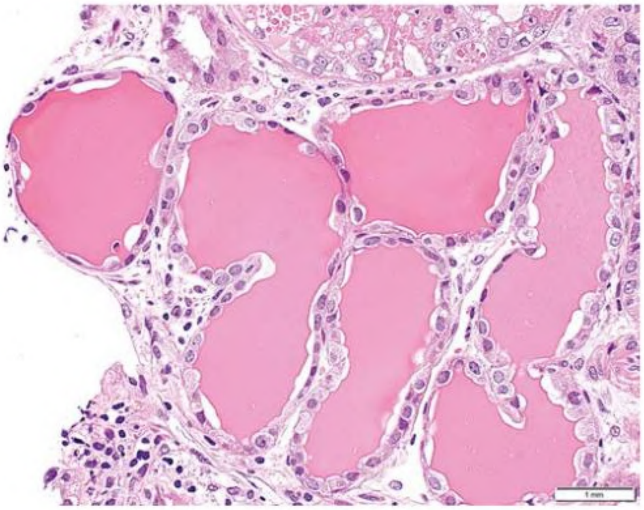

Fig. 19.9 Focal Segmental Glomerulosclerosis, Collapsing Variant. The kidney parenchyma contains abundant tubular microcysts with proteinaceous casts. (H&E; ×200.)

glomerular epithelial cells (Fig. 19.8). Studies suggest that dysregulated podocytes, activated parietal cells (expressing claudin and CD44), and progenitor cells (expressing stem cell markers CD133 and CD24) that line Bowman's capsule contribute to the glomerular epithelial cell hyperplasia and glomerulosclerosis.[55]

In collapsing FSGS, there is prominent tubulointerstitial disease, including acute tubular injury, tubular atrophy, interstitial fibrosis, interstitial edema, and inflammation. A distinctive feature is the presence of dilated tubules forming microcysts that contain proteinaceous casts (Fig. 19.9). On EM, there is typically severe foot process effacement affecting both collapsed and noncollapsed glomeruli. Collapsing glomerulopathy may occur as a primary form of FSGS.[56] It is also frequently observed in secondary FSGS caused by viral infections (including HIV and SARS-CoV-2), lupus podocytopathy, hemophagocytic syndrome, and interferon therapy in conjunction with high-risk *APOL1* variants.[29,32,57] The presence of endothelial tubuloreticular inclusions helps to identify collapsing glomerulopathy secondary to HIV-associated nephropathy, COVID-19–associated nephropathy

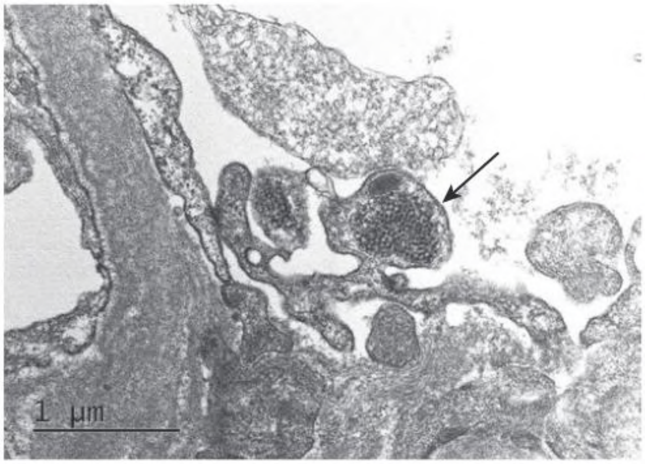

Fig. 19.10 **Focal Segmental Glomerulosclerosis, Collapsing Variant, COVID-19–associated nephropathy.** A glomerular endothelial cell contains a large intracytoplasmic tubuloreticular inclusion ("interferon footprint"; *arrow*) composed of interanastomosing tubular structures within a dilated cisterna of endoplasmic reticulum. (EM; ×40,000.)

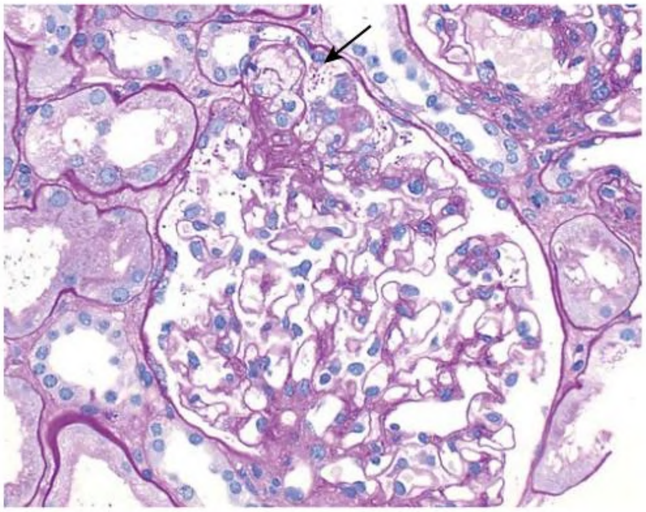

Fig. 19.11 **Focal Segmental Glomerulosclerosis, Tip Lesion Variant.** A cellular tip lesion displays engorgement of glomerular capillaries by foam cells and adhesion of the involved segment to the origin of the proximal tubule/tubular pole *(arrow).* (PAS; ×200.)

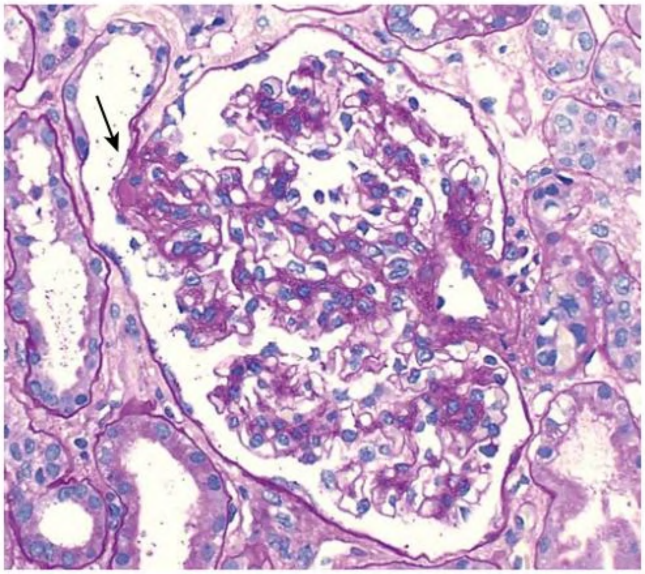

Fig. 19.12 **Focal Segmental Glomerulosclerosis, Tip Lesion Variant.** A sclerosing tip lesion forms an adhesion to the tubular pole *(arrow).* (PAS; ×200.)

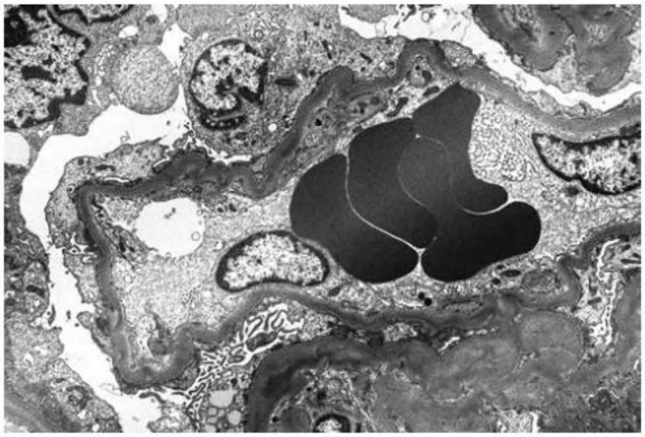

Fig. 19.13 **Focal Segmental Glomerulosclerosis, Tip Lesion Variant.** Electron microscopy reveals complete podocyte foot process effacement with matting of the actin cytoskeleton, selling of podocyte cell bodies, and focal microvillous transformation of podocyte cytoplasm. (EM; ×3000).

lupus podocytopathy, or interferon therapy. These "interferon footprints" consist of 24-nm interanastomosing tubular structures located within dilated cisternae of endoplasmic reticulum (Fig. 19.10).

Tip Variant of Focal Segmental Glomerulosclerosis

The tip variant is defined by the presence of at least one segmental lesion involving the tip domain, the outer 25% of the tuft next to the origin of the proximal tubule.[47,48,58] There is either adhesion between the tuft and Bowman's capsule or confluence of swollen podocytes with parietal or tubular epithelial cells at the tubular lumen or neck. The segmental lesions may be cellular or sclerosing (Figs. 19.11 and 19.12). The presence of perihilar sclerosis or collapsing sclerosis rules out the tip variant. In one study of FSGS tip lesions, biopsy specimens had glomerular tip lesions alone in 26% and glomerular tip lesions plus other peripheral FSGS lesions in the other 74%.[48] The degree of foot process effacement is usually severe (Fig. 19.13). Most cases are primary and resemble MCD with respect to abrupt onset of full nephrotic syndrome, high likelihood of responsiveness to corticosteroid therapy, and favorable prognosis relative to other FSGS variants. Higher shear stress and tuft prolapse at the tubular pole are likely to play a role in the morphogenesis of this lesion.[4]

Other Variants of Focal Segmental Glomerulosclerosis

Two histologic variants often included within the FSGS spectrum are FSGS with diffuse mesangial hypercellularity and C1q nephropathy (see Chapter 29).[1] FSGS with *diffuse mesangial hypercellularity* has lesions of FSGS on a background of generalized mesangial hypercellularity. By IF, there is often diffuse mesangial positivity for IgM, with more variable mesangial staining for C3. EM reveals extensive foot process effacement. Amorphous hyaline-like electron-dense deposits are variably present in the paramesangial region. This variant occurs almost exclusively in young children.[59,60]

C1q nephropathy is defined by dominant or codominant IF staining for C1q, mesangial electron-dense deposits, and light microscopic findings resembling FSGS or MCD with variable mesangial hypercellularity. In one study, 17 patients had a light microscopic appearance of

BOX 19.3 Risk Factors for Progressive Kidney Disease in Focal Segmental Glomerulosclerosis

Clinical Features at Biopsy
- Severity of nephrotic-range proteinuria
- Elevated serum creatinine
- Black race

Histopathologic Features at Biopsy
- Collapsing variant
- Tubulointerstitial fibrosis

Clinical Features During Disease Course
- Failure to achieve partial or complete remission

FSGS (including six collapsing and two cellular) and three of MCD.[53] In addition to C1q staining, biopsy specimens may show deposition of other immunoglobulins (particularly IgG) and complement components (C3), making it important to exclude other clinical disease such as lupus nephritis. In C1q nephropathy, electron-dense deposits are typically located in the paramesangial region subjacent to the GBM reflection. There is variable foot process effacement. In one series of C1q nephropathy, many cases represented a subgroup of primary FSGS or MCD, whereas in others, an idiopathic immune complex–mediated glomerulonephritis was noted.[54]

Distinguishing Pathologic Features of Secondary Focal Segmental Glomerulosclerosis

Although the pathology of some secondary forms of FSGS closely resembles primary FSGS, there are several noteworthy differences. Collapsing FSGS, particularly when tubuloreticular inclusions are present, should prompt exclusion of viral, autoimmune, or drug-induced causes of disease. In secondary adaptive forms of FSGS, kidney biopsy typically shows glomerulomegaly and predominantly perihilar lesions of segmental sclerosis and hyalinosis. In secondary FSGS resulting from loss of kidney mass, such as from reflux nephropathy or hypertensive arterionephrosclerosis, FSGS is usually seen on a background of extensive global glomerulosclerosis, tubular atrophy, interstitial fibrosis, and arteriosclerosis. In secondary FSGS related to sickle cell disease, glomerular hypertrophy and sclerosis are associated with capillary congestion by sickled erythrocytes and double contours of the GBM resembling those seen in chronic thrombotic microangiopathy.[61,62] In adaptive forms of FSGS, the degree of foot process effacement tends to be relatively mild, affecting less than 50% of the total glomerular capillary surface area. In one study, a cutoff of greater than 1500 nm for mean foot process width was able to distinguish primary from adaptive FSGS.[16]

NATURAL HISTORY AND PROGNOSIS

The natural history of FSGS is varied. Without response to therapy, patients with primary FSGS experience a progressive increase in proteinuria and progression to kidney failure. Spontaneous remission is uncommon.[63] In treatment-refractory patients, ESKD occurs 5 to 20 years from presentation, with approximately 50% of such patients developing kidney failure within 10 years.[64]

Certain epidemiologic, clinical, and histologic findings at diagnosis help predict the long-term course of patients with FSGS (Box 19.3). African Americans with FSGS experience a more rapid progression to ESKD than people of other races.[65] At biopsy, reduced GFR, greater degrees of proteinuria, and greater degrees of interstitial fibrosis predict a more progressive course.[66,67] In general, outcomes are best for tip variant and worst for collapsing variant of primary FSGS, with intermediate outcome in FSGS NOS.[47,68] In a comparative series, the percentage of complete and partial remission was greatest for tip lesion (76%), lowest for collapsing variant (13%), and intermediate for cellular (44%) and FSGS NOS (39%). There was a strong inverse correlation between remission rates and progression to ESKD among these subgroups. Accordingly, the likelihood of ESKD was greatest for collapsing variant (65%), lowest for tip lesion (6%), and intermediate for cellular variant (28%) and FSGS NOS (35%).[47] Although collapsing variant has the worst outcome among the FSGS variants, when patients with collapsing variant are matched with patients with FSGS NOS for baseline levels of kidney function, proteinuria, and immunosuppression, responses to treatment are similar, highlighting the importance of early detection and aggressive therapy.[50] A key predictor of prognosis is remission of proteinuria. Complete remission of the nephrotic syndrome is associated with excellent kidney survival compared with those who do not remit.[64] Even patients with a partial remission of nephrotic syndrome have a lower rate of long-term kidney failure.[64]

TREATMENT

Before considering the treatment of FSGS, it is useful to define remission of proteinuria. There has been considerable variability in the definition of remission, and the KDIGO Guideline has attempted to unify these definitions (Box 19.2).[11] Recently a novel definition of partial remission (40% proteinuria reduction and proteinuria <1.5 g/g) was useful in predicting prognosis.[69]

Although FSGS is common in adults, there are few randomized controlled trials (RCTs) on which to base treatment decisions.[11] However, almost all patients with FSGS will benefit from good supportive care (e.g., blockade of the renin-angiotensin-aldosterone system), as described in Chapter 82. The optimal treatment of FSGS is not well defined. In part, this relates to properly defining primary and secondary forms of the disease, including unrecognized genetic variants (Fig. 19.14). For example, even after biopsy, it is not always clear whether an obese person with glomerulomegaly, FSGS lesions, and nephrotic-range proteinuria has secondary or primary disease. This problem of heterogeneity in the patient population also potentially translates to varying treatment responses in clinical trials.[70] In general, we recommend treating patients with presumed primary FSGS with immunosuppressive medications. In contrast, immunosuppression is not needed in patients with subnephrotic proteinuria, genetic or secondary FSGS, or FSGS of uncertain etiology (Fig. 19.14).[11]

First-Line Therapy for Focal Segmental Glomerulosclerosis

Corticosteroids as Initial Therapy (Table 19.2)

Corticosteroids appear to increase the likelihood of remission[64,71-73] and are considered first-line therapies for primary FSGS.[11] Although earlier reports suggested that primary FSGS is a steroid-resistant disease with poor outcomes,[74] subsequent studies showed that longer treatment duration with higher doses of corticosteroids was associated with improved response rates of 50% to 60%,[63,66,75] and in one study up to 80%.[76] The suggested regimens (as per KDIGO Guideline) are shown in Box 19.2.[11] A suggested maximum duration of high-dose corticosteroid treatment for remission to be achieved is 16 weeks. However, based on available evidence, it is uncertain whether the side effects of 16 weeks of high-dose glucocorticoid treatment outweigh benefits in primary FSGS, as studies have been inconsistent in the reporting of adverse events. The optimal duration of glucocorticoid therapy is

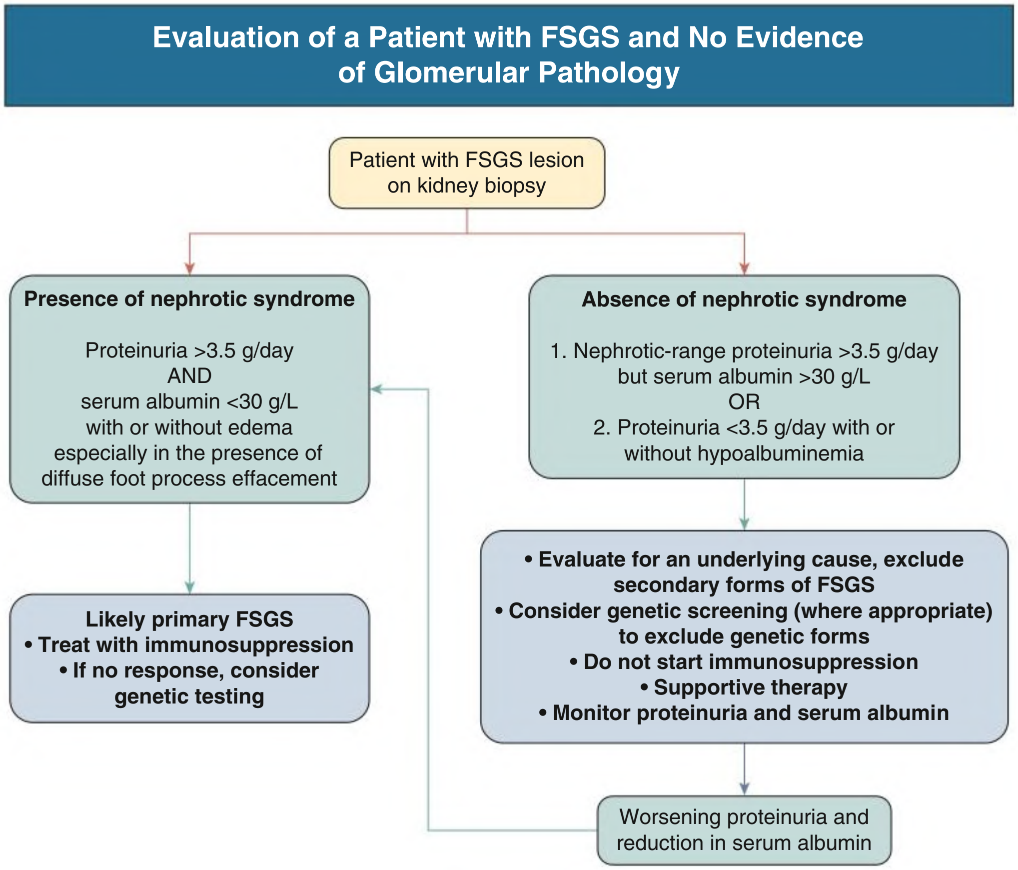

Fig. 19.14 Evaluation of a patient with focal segmental glomerulosclerosis (FSGS) lesion on the kidney biopsy and no evidence of other glomerular pathology. (From KDIGO GN Guideline. *Kidney Int.* 2021;100(4S):S1–S276.)

unknown. A slow taper is suggested after initial response, and for such patients, the suggested total period of glucocorticoid treatment is 6 months as tolerated.[11] If there is no remission in proteinuria, corticosteroids should be tapered off after 16 weeks.

Corticosteroid-Sparing or Noncorticosteroid Regimens as Initial Therapy

Patients may not tolerate high doses of glucocorticoids, and with the need for prolonged treatment regimens for FSGS, adverse effects of corticosteroids may be unacceptable (Table 19.2). Moreover, in patients with obesity, uncontrolled diabetes, psychiatric disease, or severe osteoporosis, corticosteroids may be relatively contraindicated.[77] CNIs (cyclosporine or tacrolimus) and antimetabolites (azathioprine or mycophenolic acid analogs) have been used to avoid or reduce exposure to corticosteroids. In a small observational study, tacrolimus monotherapy achieved partial remission in all six patients after about 6 months.[78] In a randomized study of 33 FSGS patients, mycophenolate mofetil (MMF) 2 g/day for 6 months with prednisolone 0.5 mg/kg/day for 2 to 3 months was compared with conventional therapy with prednisolone 1 mg/kg/day for 3 to 6 months. Remission was similar in the MMF group 70% versus 69% in the conventional corticosteroid group.[79] A retrospective observational study compared high-dose oral prednisone (1 mg/kg/day) for at least 4 months followed by a taper, with lower dose prednisone (0.5 mg/kg/day) in combination with cyclosporine (3 mg/kg/day initially, tapering to 50 mg/day) or azathioprine (2 mg/kg/day initial dose, tapering to 0.5 mg/kg/day). Average duration of treatment was 20 months. Low-dose prednisone was given to 16 patients with obesity, bone disease, or mild diabetes. Remission rates were comparable: 63% for prednisone, 80% for prednisone plus azathioprine, and 86% for prednisone plus cyclosporine.[72] In a large retrospective study, 173 patients were treated with corticosteroids alone and 90 treated with CNIs with or without corticosteroids; the risk of ESKD was similar in the two groups.[80]

Corticosteroid-Resistant Focal Segmental Glomerulosclerosis

For patients with corticosteroid-resistant FSGS, the CNIs (cyclosporine or tacrolimus) are recommended.[11] Other options include MMF, rituximab, and corticotrophin.

Calcineurin Inhibitors (Table 19.3)

In one study, steroid-resistant adult patients with FSGS randomized to either cyclosporine with low-dose corticosteroids or the same low dose of corticosteroids alone for a 6-month period exhibited a much higher remission rate (12% complete and >70% complete or partial) in the group treated with cyclosporine.[81] Despite some relapses after cyclosporine discontinuation, the treated group had significantly more patients in remission and better GFR at long-term follow-up. In another randomized controlled study, cyclosporine monotherapy was used, with similar results.[82] Although data on tacrolimus are more limited, results are similar.[83] Because the major side effects (nephrotoxicity, hypertension, and hyperkalemia) are similar with cyclosporine and tacrolimus, the choice depends on other adverse effects (e.g., gum swelling, tremor, hirsutism with cyclosporine, and diabetes with tacrolimus). Because there is a high relapse rate when cyclosporine is discontinued early,[81] we use a 1-year treatment course with a slow taper in patients with a favorable reduction of proteinuria on CNIs.[11]

TABLE 19.2 Initial Treatment of Primary Focal Segmental Glomerulosclerosis

Treatment	Dose and Duration
Glucocorticoids	**Starting Dose**
	• High-dose glucocorticoid therapy with prednisone at daily single dose of 1 mg/kg (maximum 80 mg) or alternate-day dose of 2 mg/kg (maximum 120 mg)
	High-Dose Glucocorticoid Treatment Duration
	• Continue high-dose glucocorticoid therapy for at least 4 weeks and until complete remission is achieved, or a maximum of 16 weeks, whichever is earlier
	• Patients who are likely to remit will show some degree of proteinuria reduction before 16 weeks of high-dose treatment
	• It may not be necessary to persist with high-dose glucocorticoid therapy until 16 weeks if the proteinuria is persistent and unremitting, especially in patients who are experiencing side effects
	Glucocorticoid Tapering
	• If complete remission is achieved rapidly, continue high-dose glucocorticoid treatment for 2 weeks after the disappearance of proteinuria, whichever is longer. Reduce prednisone by 5 mg every 1–2 weeks to complete a total duration of 6 months.
	• If partial remission is achieved within 8–12 weeks of high dose glucocorticoid treatment, continue until 16 weeks to ascertain whether further reduction of proteinuria and complete remission may occur. Thereafter, reduce the dose of prednisone by 5 mg every 1–2 weeks to complete a total duration of 6 months.
	• If the patient proves to be steroid resistant or develops significant toxicities, glucocorticoids should be rapidly tapered as tolerated and treatment with alternative immunosuppression like a CNI inhibitor should be considered
CNIs	**Starting Dose**
	• Cyclosporine 3–5 mg/kg/day in two divided doses *or* tacrolimus 0.05–0.1 mg/kg/day in two divided doses
	• Target trough levels could be measured to minimize nephrotoxicity
	• Cyclosporine target trough level: 100–175 ng/mL
	• Tacrolimus target trough level: 5–10 ng/mL
	Treatment Duration for Determining CNI Efficacy
	• Cyclosporine or tacrolimus should be continued at doses achieving target trough level for at least 6 months, before considering the patient to be resistant to CNI treatment
	Total CNI Treatment Duration
	• In patients with partial or complete remissions, cyclosporine or tacrolimus should be continued at doses achieving target trough level for at least 12 months to minimize relapses
	• The dose of cyclosporine or tacrolimus can be slowly tapered over a course of 6–12 months as tolerated
	• Consider discontinuing CNI if the eGFR continues to decline to <30 mL/min/1.73 m^2

CNI, Calcineurin inhibitor; *eGFR,* Estimated glomerular filtration rate.
Modified from KDIGO GN Guideline. *Kidney Int.* 2021;100(4S):S1–S276.

TABLE 19.3 Treatment of Steroid-Resistant Primary Focal Segmental Glomerulosclerosis

Treatment	Dose and Duration
CNIs	**Starting Dose**
	• Cyclosporine 3–5 mg/kg/day in two divided doses *or* tacrolimus 0.05–0.1 mg/kg/day in two divided doses
	• Target trough levels could be measured to minimize nephrotoxicity
	• Cyclosporine target trough level: 100–175 ng/mL
	• Tacrolimus target trough level: 5–10 ng/mL
	Treatment Duration for Determining CNI Efficacy
	• Cyclosporine or tacrolimus should be continued at doses achieving target trough level for at least 6 months before considering the patient to be resistant to CNI treatment
	Total CNI Treatment Duration
	• In patients with partial or complete remissions, cyclosporine or tacrolimus should be continued at doses achieving target trough level for at least 12 months to minimize relapses
	• The dose of cyclosporine or tacrolimus can be slowly tapered over a course of 6–12 months as tolerated
	• Consider discontinuing CNI if the eGFR continues to decline to <30 mL/min/1.73 m^2
Inability to tolerate or contraindications to CNIs	• Lack of quality evidence for any specific alternative agents
	• MMF and high-dose dexamethasone, rituximab, and ACTH have been considered
	• Treatment will need to be personalized and is dependent on availability of drugs and resources, as well as the benefits of further treatment and risks of adverse effects of immunosuppression
	• Patients should be referred to specialized centers with the appropriate expertise and should be evaluated on the appropriate use of alternative treatment agents or to discontinue further immunosuppression

ACTH, Adrenocorticotrophic hormone; *CNI,* calcineurin inhibitor; *eGFR,* estimated glomerular filtration rate; *MMF,* mycophenolate mofetil.
Modified from KDIGO GN Guideline. *Kidney Int.* 2021;100(4S):S1–S276.

Mycophenolic Acid Analogs

Mycophenolic acid analogs (mycophenolate-MMF and mycophenolic acid) have been used successfully in several uncontrolled series of patients with FSGS.[84,85] One multicenter prospective RCT of cyclosporine versus oral MMF plus dexamethasone in 138 children and adults up to age 40 years who had corticosteroid-resistant FSGS showed no difference in remission rates.[86] However, this trial did not meet recruitment targets and was likely underpowered. Further, as with other clinical trials of FSGS, patients with secondary FSGS were likely included.

Other Therapies

Use of sirolimus has been limited since the drug was reported to induce proteinuria and FSGS lesions in kidney transplant recipients. In one series, sirolimus was associated with worsening of kidney function, episodes of acute kidney injury, and no remissions of the nephrotic syndrome.[87] The addition of plasmapheresis to immunosuppressive medications, which has been successful in treating some patients with recurrent FSGS in the kidney allograft (see Chapter 113), has not been evaluated in patients with native kidney FSGS. Likewise, use of low-density lipoprotein apheresis and galactose (a monosaccharide sugar with a high affinity for CLC1, a putative permeability factor in patients with FSGS) has been anecdotal.[88,89] Rituximab has been used anecdotally in several studies of small numbers of patients with FSGS who have either not responded to other treatments or have become dependent on these therapies. It has proved more successful for steroid-dependent[90] than steroid-resistant patients.[91] Corticotropin (adrenocorticotropic hormone) has shown benefit in small numbers of patients with FSGS resistant to multiple other immunosuppressive agents, but it also has not been studied in large RCTs.[92]

Therapy of Secondary Focal Segmental Glomerulosclerosis

For patients with secondary forms of FSGS, treating the underlying cause should be the initial step. Patients with FSGS secondary to obesity and heroin nephropathy have had remissions of proteinuria after weight reduction or cessation of heroin use, respectively. In HIV-associated nephropathy, therapy with highly active antiretroviral drugs and blockers of the renin-angiotensin system has proven useful (see Chapter 58). The role of immunosuppression has not yet been proven in RCTs in any form of secondary FSGS. In all forms of secondary FSGS, supportive therapy as outlined in Chapter 82 is essential to prevent progressive kidney disease. In patients with primary idiopathic or a secondary form of FSGS who remain nephrotic, fluid retention and edema can be managed with salt restriction and diuretics (see Chapter 16). In general, genetic forms of FSGS are not responsive to immunosuppressive medications. Although blockers of the renin-angiotensin system are commonly used, there are no studies that have reported outcomes. A small proportion of children with genetic FSGS have demonstrated proteinuria reduction with CNIs.[93,94]

KIDNEY TRANSPLANTATION

Approximately 30% of patients with primary FSGS who develop ESKD and undergo kidney transplantation develop recurrent FSGS in the allograft. Risk factors for recurrence include early-onset FSGS, and severe proteinuria with a rapid course to ESKD. Patients who have lost a prior allograft due to recurrent FSGS are at highest risk for recurrence. Treatment protocols include plasmapheresis, rituximab, high-dose CNIs, and renin-angiotensin system blockers.[95] Recurrent FSGS is further discussed in Chapter 113.

SELF-ASSESSMENT QUESTIONS

1. Use of which of the following medications has *not* been associated with proteinuria and the histologic pattern of FSGS?
   A. Interferon β
   B. Pamidronate
   C. Sirolimus
   D. Amlodipine
   E. Lithium
2. Which histologic feature on kidney biopsy suggests a secondary FSGS pattern related to structural-functional adaption?
   A. Adhesions or synechiae to Bowman's capsule
   B. Glomerulomegaly and perihilar segmental sclerosis and hyalinosis
   C. Hypertrophy and hyperplasia of the visceral epithelial cells
   D. Diffuse mesangial hypercellularity
   E. Collapse of part of or all the glomerular tuft
3. Which of the following statements about the prognosis of primary FSGS is *true?*
   A. The outcome is worse for patients with the tip variant versus other patterns.
   B. The outcome is better for patients with the collapsing variant versus other patterns.
   C. Remission of proteinuria is a strong predictor of a favorable outcome.
   D. African Americans have a prognosis similar to that of Whites if controlling for hypertension and degree of proteinuria.
   E. The degree of glomerular sclerosis is the strongest histologic predictor of a poor kidney outcome.

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